Rachit Kuklani

(716) 709-0483 rachitta@buffalo.edu LinkedIn Portfolio GitHub LeetCode

EDUCATION

University at Buffalo Buffalo, NY
B.Sc. (Honors Specialization) in Computer Science *Expected May 2028*
• GPA: 3.91/4.0 — Dean's Honor List (Spring 2025-26)

EXPERIENCE

Software and Data Analytics Intern May 2026 – Aug 2026
Biz Analytix, New York City **Tech:** ASP.NET MVC, React, .NET Framework, JavaScript, SQL Server, LINQ

- Built responsive analytics pages in **React** and used .NET Framework and MVC architecture to build a service layer on the Back end that returns database driven responses replacing legacy server related workflows with reusable API driven components reducing development time by **40%**.
- Refactored and optimized SQL and LINQ queries, replacing inefficient data-access logic with streamlined filtering and aggregation to accelerate data retrieval and improve React page-rendering speed by **65%**.

Algorithms and Complexity — Teaching Assistant Aug 2025 – Present
University at Buffalo – School of Engineering and Applied Sciences **Tech:** Python, Java, C, NumPy, MySQL

- Debugged and optimized student implementations of complex algorithms (graph theory, dynamic programming), leveraging NumPy vectorization to help students achieve **2–5× faster runtimes**.
- Conducted 60+ in-depth **C/C++** code reviews, mentoring students on pointer arithmetic, manual memory management, and POSIX threads; automated grading scripts and testing workflows reduced grading time per submission by ∼**70%**, enabling **95%** of mentees to achieve zero memory leaks.

Software Engineering Intern May 2025 – Aug 2025
Kuksons Electronics, Mumbai, India **Tech:** AWS RDS, React, Node.js, SCADA, SIEMENS TIA Portal, PostgreSQL

- Developed a responsive frontend dashboard using React to monitor project status and inventory levels for ongoing automation tasks, enabling real-time visibility for engineers and reducing manual status tracking by **40%**.
- Built a centralized PLC error-logging pipeline using **Python, PostgreSQL, and AWS RDS** to capture and persist runtime faults across production lines, reducing troubleshooting time by **35%** and enabling root-cause analysis of **40+ daily errors**.

PROJECTS

Dual-Dimension Graphics Engine (DDGE) [GitHub] *OOP, C++, Java, Box2D, Vulkan, Git*

- Engineered a high-performance Vulkan pipeline delivering 60+ FPS in multi-object scenes, achieving **20–40%** faster frame preparation than typical student frameworks (OpenGL, SFML, SDL).
- Engineered a Vulkan pipeline delivering **60+ FPS** and **20–40%** faster frame preparation in multi-object scenes.

Instant Messenger (RESQME) [GitHub] *C, Linux/Unix, Sockets, Serialization, Memory Management, Multithreading*

- Implemented packet encoding/decoding in **C** using bit manipulation and pointer arithmetic, serializing data into compact byte streams and reducing memory overhead by approximately **40%**.
- Implemented packet encoding/decoding in **C** using bit manipulation and pointer arithmetic, serializing data into compact byte streams and reducing memory overhead by ∼**40%**.

PATENT

AI-Driven Web Content Personalization System (Patent) [Patent]
React.js, TypeScript, PostgreSQL, Pandas

- Designed data storage pipelines and responsive React/TypeScript UI for AI-driven personalization, reducing data-processing latency by ∼**35%** and increasing session engagement by ∼**27%**.

ACHIEVEMENTS

- Solved **800+** data-structures and algorithms problems on LeetCode using Java, C, and SQL, strengthening problem-solving skills across graphs, dynamic programming, trees, and binary search.

SKILLS & INTERESTS

Languages: Java, C, C++, Python, JavaScript, TypeScript, SQL Server, React, ASP.NET MVC
Frameworks/Libraries: React.js, Node.js, Pandas, NumPy, Box2D, .NET Framework
Systems/Tools: AWS RDS, Linux/Unix, Git, MySQL, PostgreSQL, Vulkan, Siemens TIA Portal, SCADA
Concepts: Object Oriented Design, DSA, Client-Server Architecture, REST API Design, Data Modeling and Validation.